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Question Paper Code : 13639

B.E. / B.TECH DEGREE EXAMINATION, NOVEMBER/ DECEMBER 2017

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

13CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2013)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. List the steps needed to execute the machine instruction ADD LOCA,R1 in terms of transfers between the components.
2. State the need for overlapping instructions.
3. Compare hardwired control and micro programmed control.
4. Why is the Wait-For-Memory- Function- Completed (WMFC) step needed when reading from or writing to main memory?
5. When do you say side effect occurs when executing an instruction?
6. What is the use of forwarding in pipelined architecture?
7. What is the purpose of Dirty/Modified bit in cache memory?
8. Distinguish between memory-mapped and program-controlled I/O.
9. What do you know about shadow registers?
10. What is an Arbiter Circuit?

PART B – (5 × 16 = 80 Marks)

11. a) i. Perform **8 X -5** using Booth's Algorithm. (10 Marks)

11. a) ii. Discuss in detail about the design of Full Adder. (6 Marks)

OR

11. b) i. With neat diagram explain Straight line sequencing and branching. (6 Marks)

11. b) ii. Compute **23/8** using non restoring division algorithm. (10 Marks)

12. a) i. List the sequence of actions needed to execute the following Instructions in terms of data path components

- **Store R5,10(R8)**
- **SUB R3,R4,R5** (10 Marks)

12. a) ii. Discuss the micro-programmed control technique for generating the control signals with suitable diagrams. (6 Marks)

OR

12. b) i. Write and explain the sequence of actions for processing the instruction **AND 50(R5), R3** in a CISC Style processor. (6 Marks)

12. b) ii. At the time the instruction **Load R6, 1000(R9)** is fetched, R6 and R9 contain the values 4200 and 85320 respectively. Memory location 5200 and 86320 contain 84500 and 75900 respectively. Write the Register transfer notation and Show the contents of the interstage registers in data path during each of the 5 execution steps of this instruction. (10 Marks)

13. a) i. Explain with a neat sketch about the different Inter stage Buffers for a 5 stage pipelined RISC style processor. (8 Marks)

13. a) ii. Consider the pipelined execution of the following sequence of instructions:

Add R4, R3, R2
OR R7, R6, R5
Subtract R8, R7, R4

Initially, registers R2 and R3 contain 15 and 20, respectively. Registers R5 and R6 contain 128 and 20, respectively. Assume that the pipeline provides forwarding paths to the ALU from registers RY and RZ. The first instruction is fetched in cycle 1, and the remaining instructions are fetched in successive cycles.

Draw a diagram to show the pipelined execution of these instructions assuming that the processor uses operand forwarding. Then, describe the contents of registers RY and RZ. (8 Marks)

OR

13. b) i. Consider the following instructions at the given addresses in the memory

1000 Add R3, R2, #200

1004 Add R5, R4, #35

1008 And R6, R4, #0xB7

1012 Subtract R7, R2, R4

Initially, registers R2 and R4 contain 2000 and 3000, respectively. These instructions are executed in a computer that has a four-stage pipeline. The first instruction is fetched in clock cycle 1, and each fetch takes 2 clock cycles, the remaining instructions are fetched in successive cycles. Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8 and show the contents of various registers during cycles 1 to 8. (8 Marks)

13. b) ii. What is Hazard? What are the various types of Hazard? How the Pipeline helps in executing the instruction if the value of source operand is not available when needed. Explain in detail with suitable example. (8 Marks)

14. a) i. Design a 64-megabit DRAM chip by organizing it as 8M x 8. Draw the internal organization of the chip and explain its working principle. (10 Marks)

14. a) ii. Explain how mapping is done between cache and main memory using direct Mapping. (6 Marks)

OR

14. b) i. A block-set-associative cache consists of a total of 512 blocks divided into 8-block sets. The main memory contains 16384 blocks, each consisting of 256 words.

A) How many bits are there in a main memory address?

B) How many bits are there in each of the TAG, SET and WORD fields?

Also, explain direct mapping techniques for the above memory specifications. How many bits are there in each of the TAG and WORD fields if an associative mapping is employed? (10 Marks)

14. b) ii. Discuss the use of DMA controllers in a computer system with neat diagram. (6 Marks)

15. a) i. Explain in detail about input process and output processes through program controlled I/O with routines. (8 Marks)

15. a) ii. Discuss in detail about Peripheral Component Interconnect. (8 Marks)

OR

15. b) i. With a neat sketch explain the function of Synchronous bus. What are the limitation of synchronous bus and state the remedial approach for it. (8 Marks)

15. b) ii. Discuss with a neat diagram about Bus Arbitration. (8 Marks)